Public Policy 2026-2027

Lecture 5: Efficiency

Thursday, October 26, 2026
Stockholm School of Economics in Riga

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What You Will Learn in Part II

1. Public policy involves government interference with private choice. What are the rationales that justify such interference?
2. Efficiency as a fundamental normative rationale for public policy.
3. But of course, efficiency is not the only value that matters for public policy. How does it relate to other important social values, like e.g. fairness?
4. Private choice ("the market") often fails to deliver socially appealing outcomes. But so does collective choice ("the government"): hence, how does the possibility of government failure inform public policy?
5. Are there generic policy instruments that are suitable to tackle market *and* government failure jointly?

What Is Efficiency? How Does It Matter in Public Policy?

Allocating Scarce Healthcare Resources

A quick classroom exercise

You are a doctor who runs a hospital in an isolated rural area. You have 20 critically ill patients: 10 patients have Disease A and 10 patients have Disease B. All the patients will die unless they quickly receive the correct dose of an expensive life-saving medication.

- Each patient with Disease A needs **one** dose of the medication to be cured.
- Each patient with Disease B needs **two** doses of the medication to be cured.

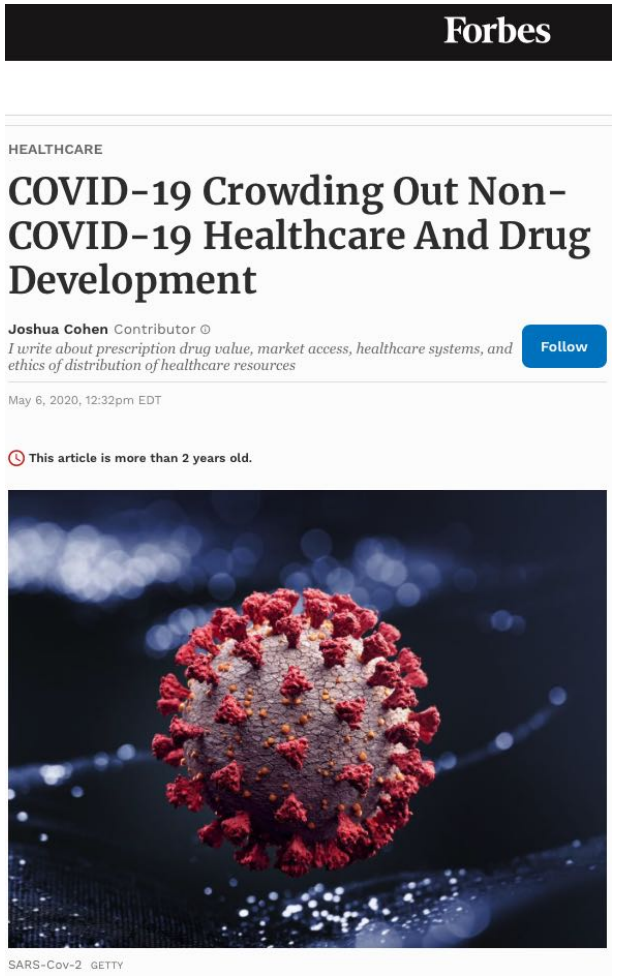
Problem: You currently have only **10 doses** of the medication, and you have neither the time nor the budget to get more.

Efficiency: A Key Normative Standard in Public Policy

Efficiency (generic definition)

Using the least amount of input to achieve the highest amount of output ("doing things right").

- Contrast with effectiveness ("doing the right things").
- What it is that some course of action is supposed to be efficient *at*?
  - 'Economic' (output: social welfare, social surplus…)
  - 'Medical' (output: # of lives saved, QALYs…)
- Economic efficiency usually the primary (often unstated) normative benchmark in public policy:
  - How can government best employ its limited resources?
  - Given a choice, the preferred course of action is the one that contributes most to the maximisation of social welfare.



Forbes

HEALTHCARE

COVID-19 Crowding Out Non-COVID-19 Healthcare And Drug Development

Joshua Cohen Contributor

I write about prescription drug value, market access, healthcare systems, and ethics of distribution of healthcare resources

Follow

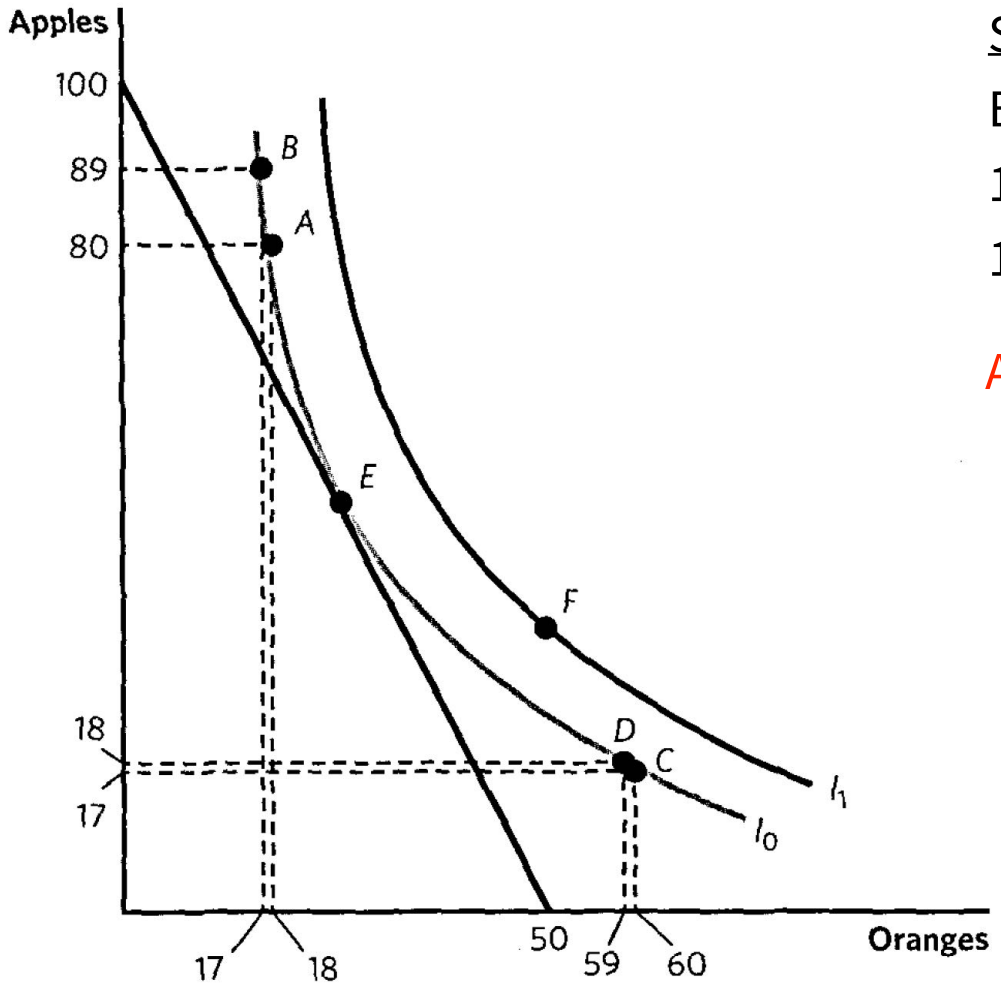
May 6, 2020, 12:32pm EDT

This article is more than 2 years old.

SARS-Cov-2 GETTY

Utilitarianism: The Roots of the Efficiency Principle in Public Policy

Optimising Private Choice Under Constraint



<u>Suppose:</u>

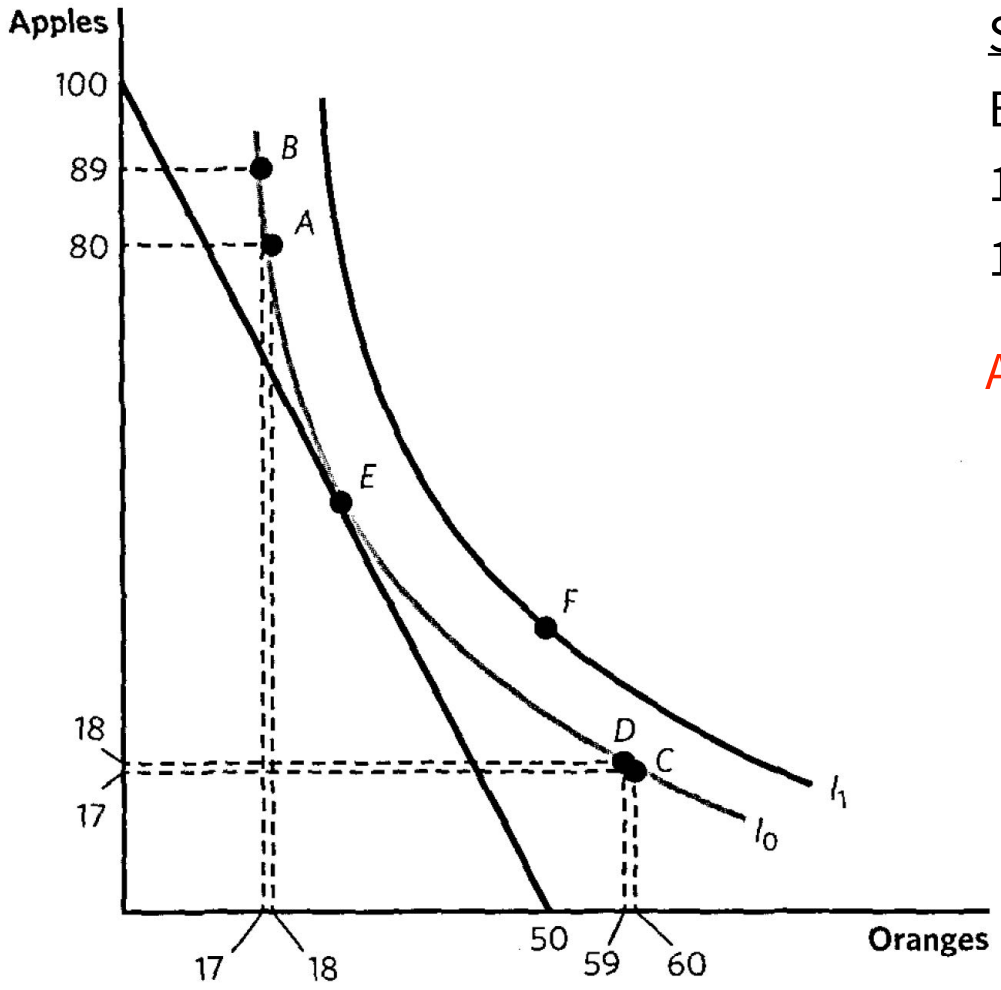
Budget: $100

1 Apple: $1

1 Orange: $2

At which point does the **consumer** achieve maximum utility?

Optimising Private Choice Under Constraint



Suppose:

Budget: $100

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1 Orange: $2

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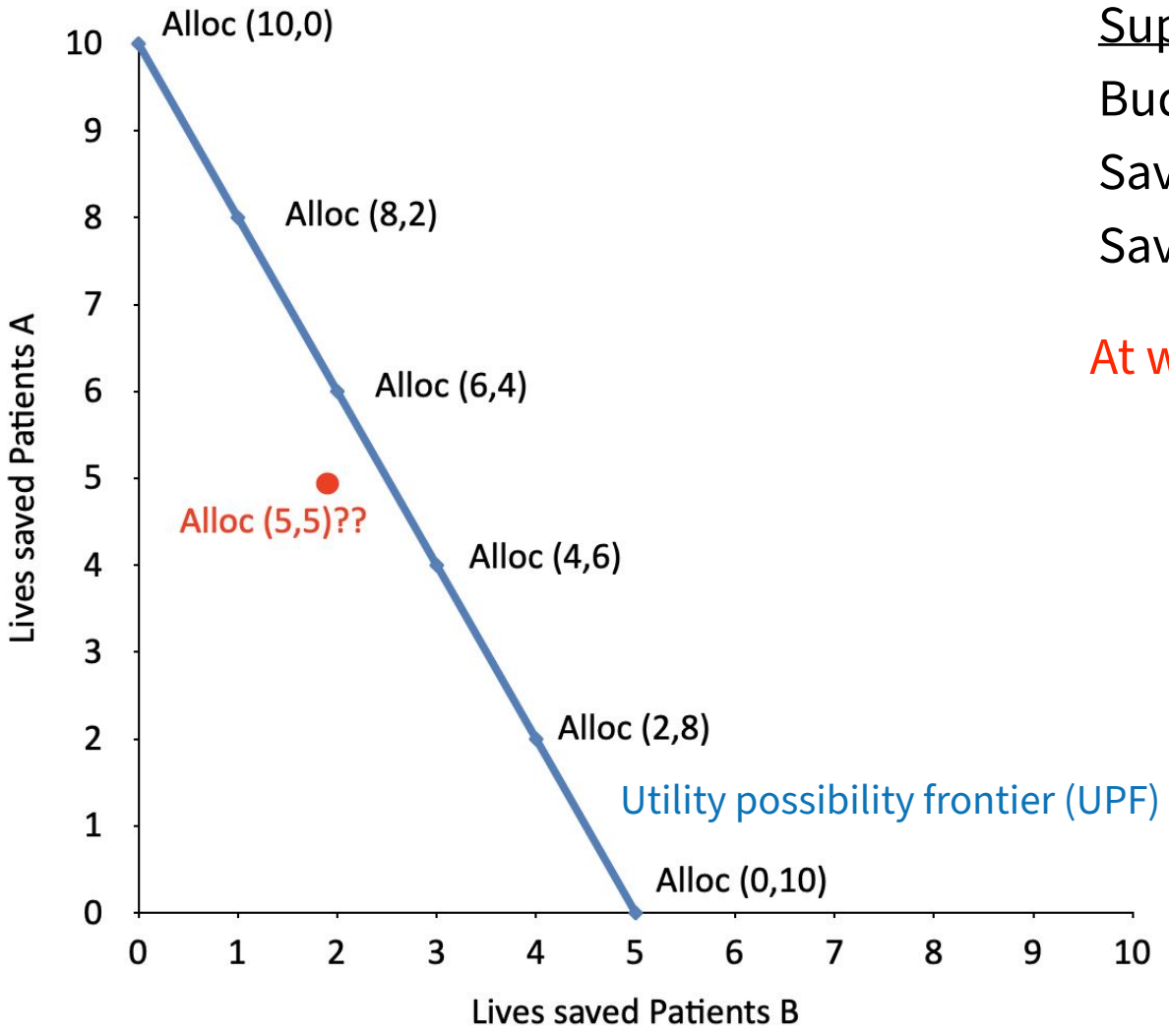
Individual utility maximised at point E, where:

$$MRS_{AO} = P_O/P_A = -2$$

Preference axioms

1. Preferences are complete
2. Preferences are transitive
3. "More is always better than less" (non-satiation)

Optimising Social Choice Under Constraint



Suppose:

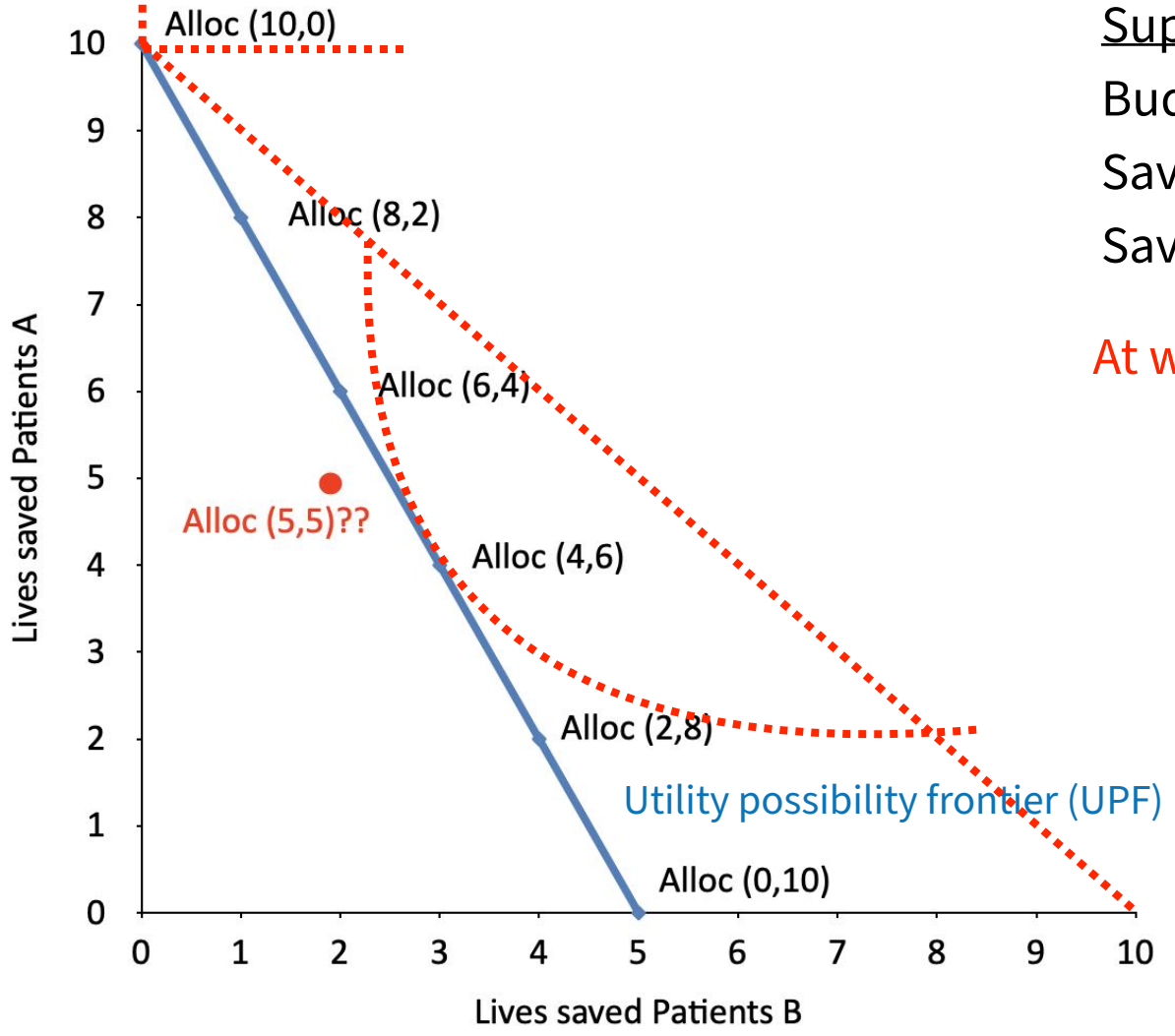
Budget: 10 doses

Saving Patient A: 1 dose

Saving Patient B: 2 doses

At which point does **society** achieve maximum utility?

Optimising Social Choice Under Constraint



Suppose:

Budget: 10 doses

Saving Patient A: 1 dose

Saving Patient B: 2 doses

At which point does **society** achieve maximum utility?

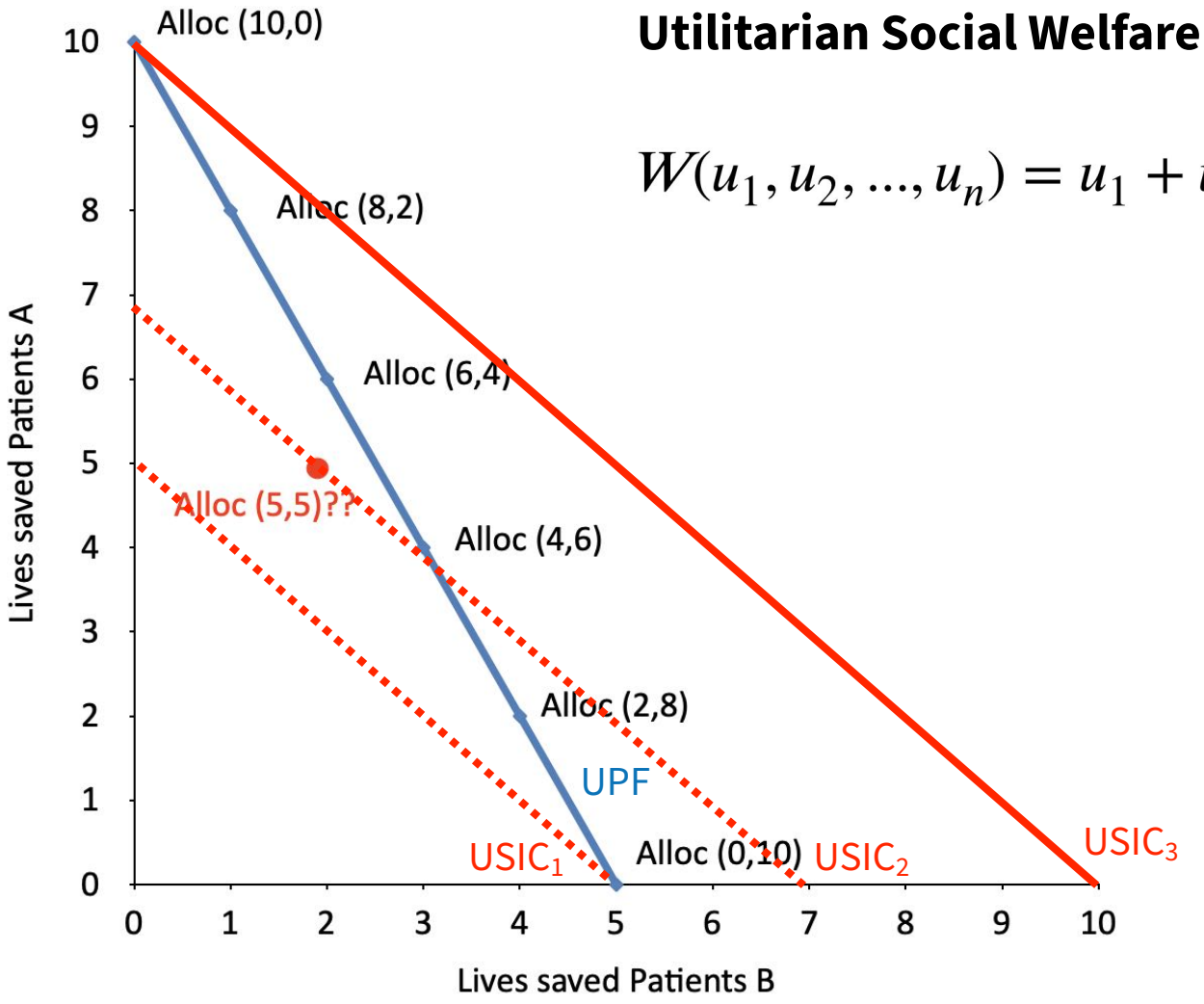
Social utility (= social welfare) maximised where:

$$SMRS_{AB} = U_B/U_A = -2$$

Social indifference curves

$SMRS_{AB}$ = the rate at which society is willing to trade off the utility of individual B for the utility of individual A while keeping **overall social welfare** constant.

Utilitarian Social Indifference Curves



Utilitarian Social Welfare Function (SWF):

$$W(u_1, u_2, ..., u_n) = u_1 + u_2 + ... + u_n = \sum_{i=1}^{n} u_i$$

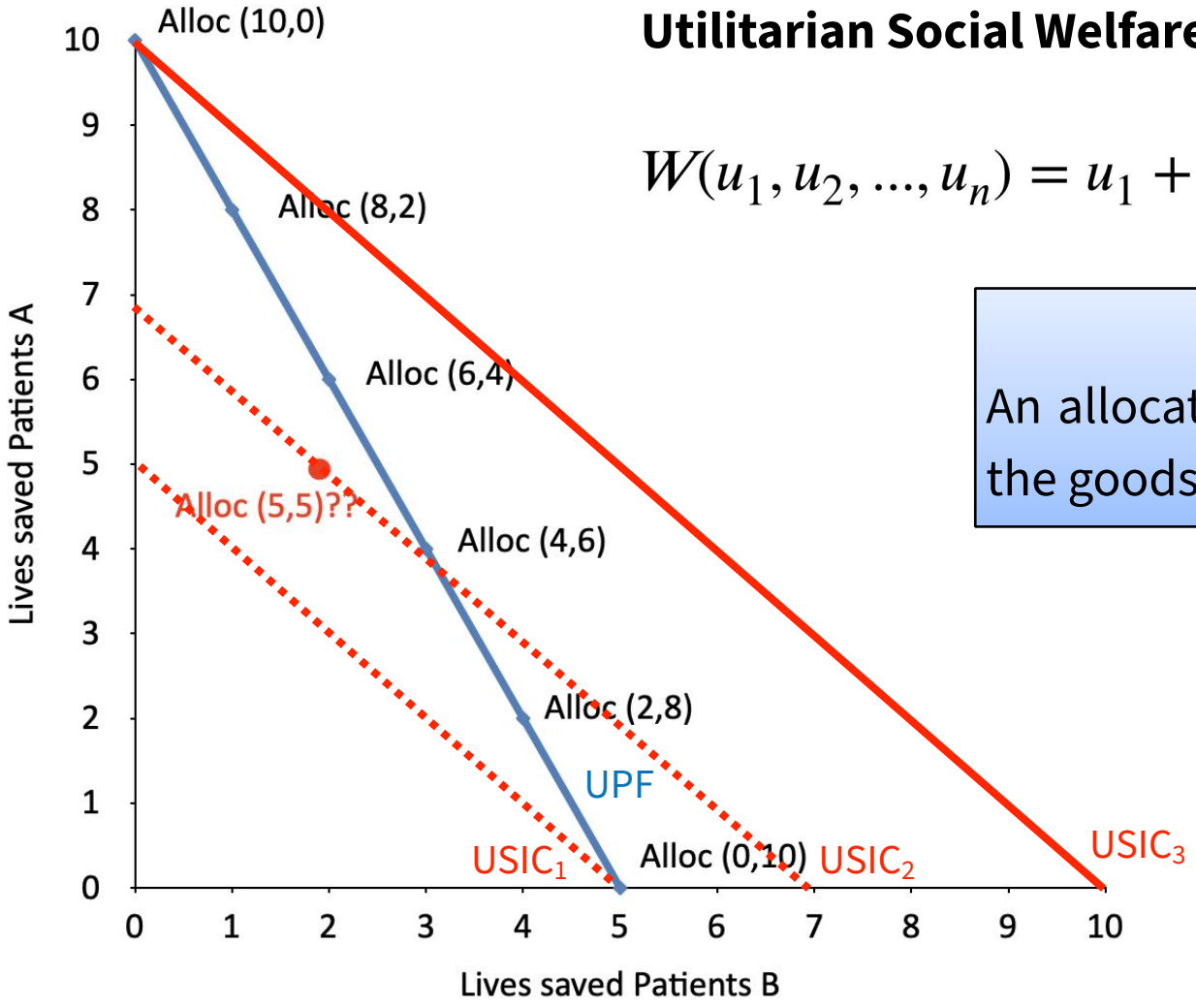


Jeremy Bentham (1748-1832)

Utilitarian Social Indifference Curves



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Utilitarian Social Welfare Function (SWF):

$$W(u_1, u_2, ..., u_n) = u_1 + u_2 + ... + u_n = \sum_{i=1}^{n} u_i$$

Utilitarian optimum:

An allocation of goods is optimal if it is impossible to reallocate the goods so as to increase the total sum of happiness in society.

Key policy implication:

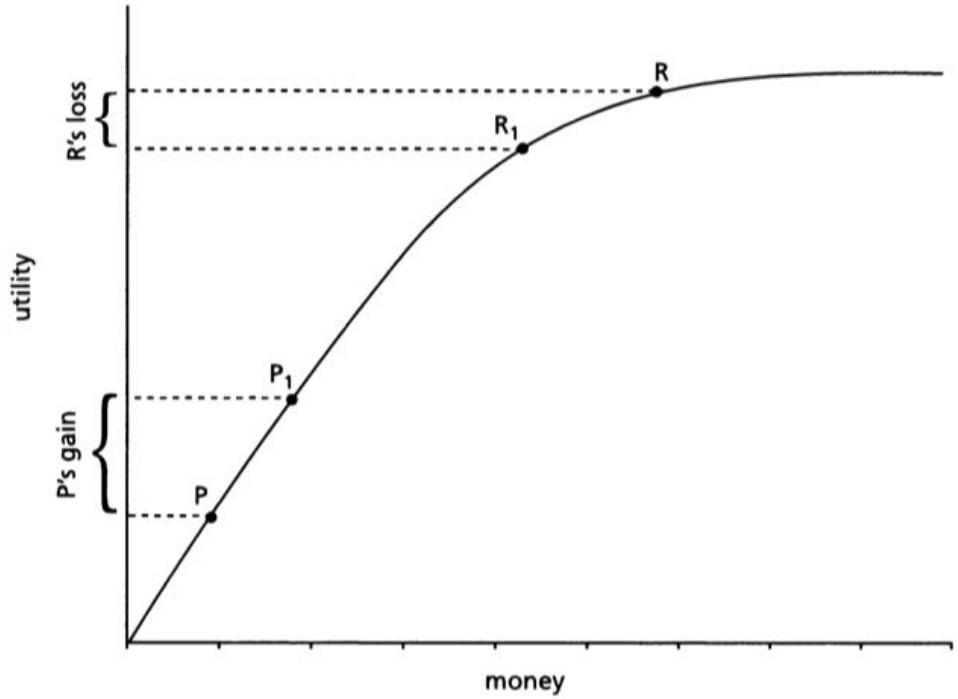
The right policy is the one that produces the "greatest good for the greatest number." Society is indifferent about even extremely unfair distributions if that is necessary for the maximisation of total happiness! (No inequality aversion)

The Trouble With Utilitarianism in Public Policy

1. Reaching the utilitarian optimum may justify doing harm:
   - It may be appropriate to sacrificing a small # of people for the sake of the greater good.
   - The aggregation of enough utility gains (however small) will eventually outweigh a small number of very large losses.
2. Utilitarianism accommodates offensive tastes: some people get satisfaction from others' disadvantage. Utilitarianism counts that satisfaction as a benefit.
3. Utilitarianism insufficiently captures special relationships:
   - Obligations towards family members, employers, kin, ethnicity, culture…?
   - Contractual agreements and promises (e.g. loans)?

The Trouble With Utilitarianism in Public Policy

4. Measurement problems: utilitarianism requires cardinal, interpersonally comparable utility. What, then, is the "currency" of utility? Money is a tricky proxy metric due to the diminishing marginal utility of money:



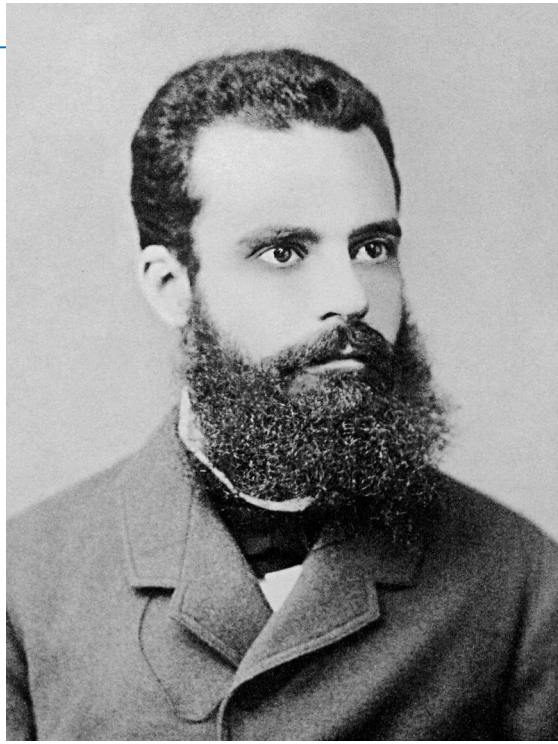
Towards a Less Controversial Version of Utilitarianism? The Pareto Framework

The “Ordinalist” Revolution of the 1920s-1930s

The transition towards contemporary welfare economics

A radical redefinition of the central point of economics:

1. Interpersonal comparisons of utility and cardinal measurements of utility no longer considered acceptable (see especially: Lionel Robbins. 1932. *An Essay on the Nature and Significance of Economic Science*. Macmillan.)
2. Positivist scientific method.
3. Farewell to substantive conceptions of utility/ welfare/ happiness; welfare to be measured ordinally by preference satisfaction.
4. Principle of minimal benevolence: If Y is better for someone than X, and X is not better for anyone than Y, then, other things being equal, Y is morally better than X.



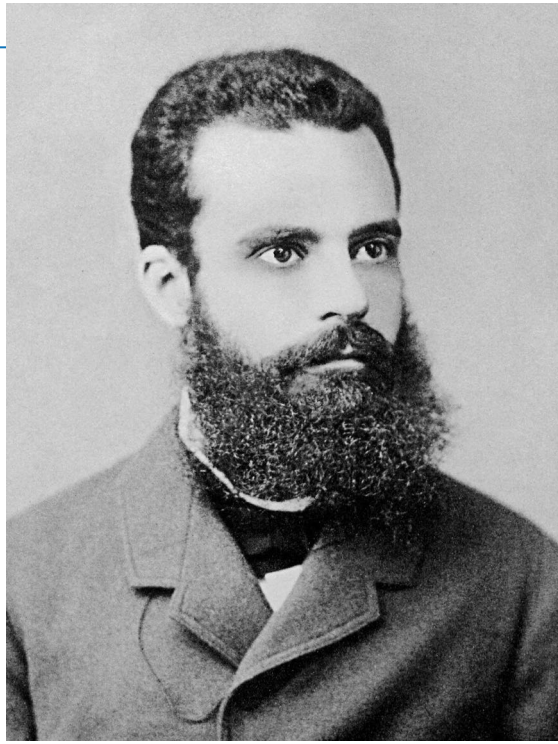
Vilfredo Pareto (1848-1923)

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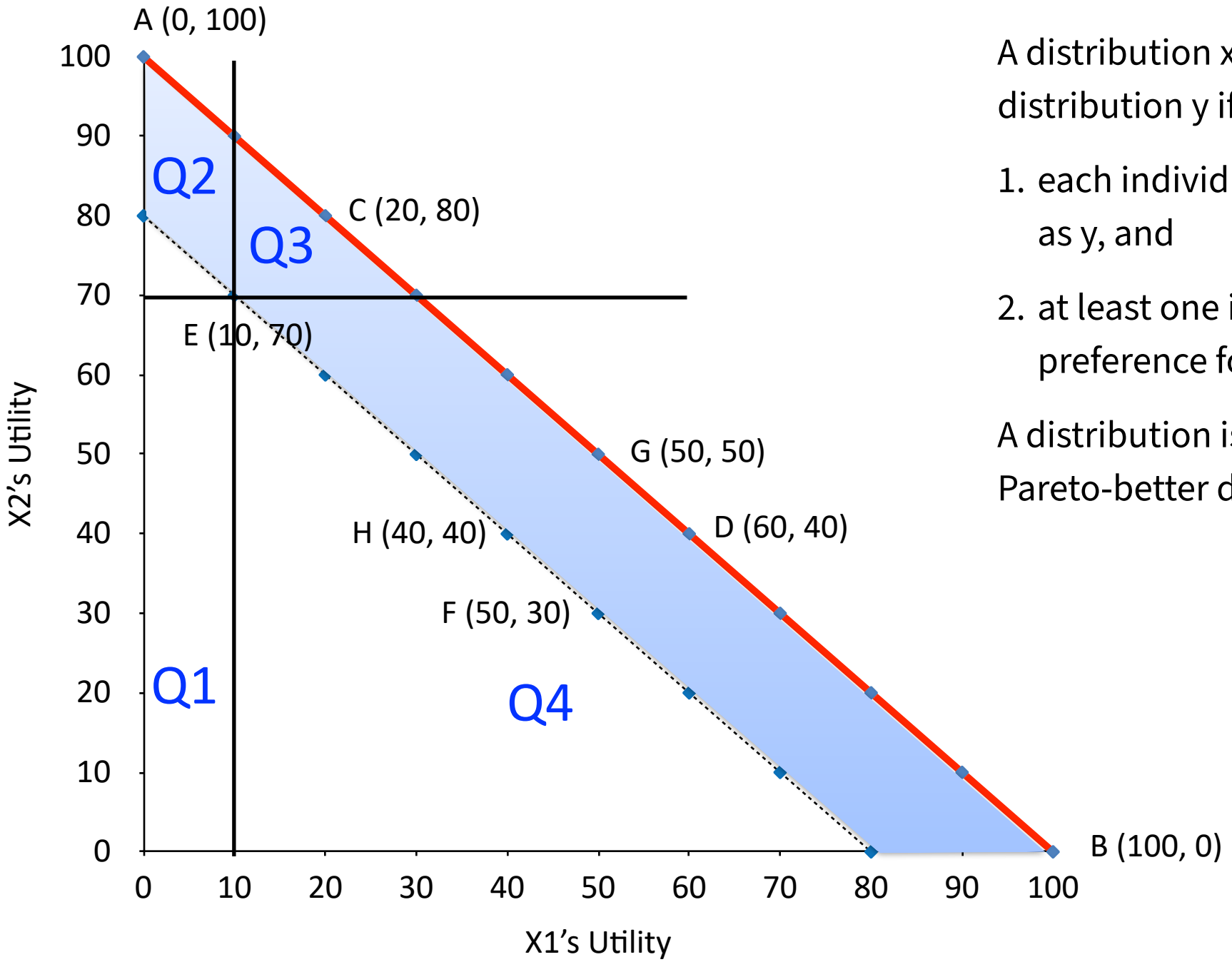
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Pareto Improvement:

A policy change that makes at least one person better off without making any other person worse off.

Pareto Efficiency:

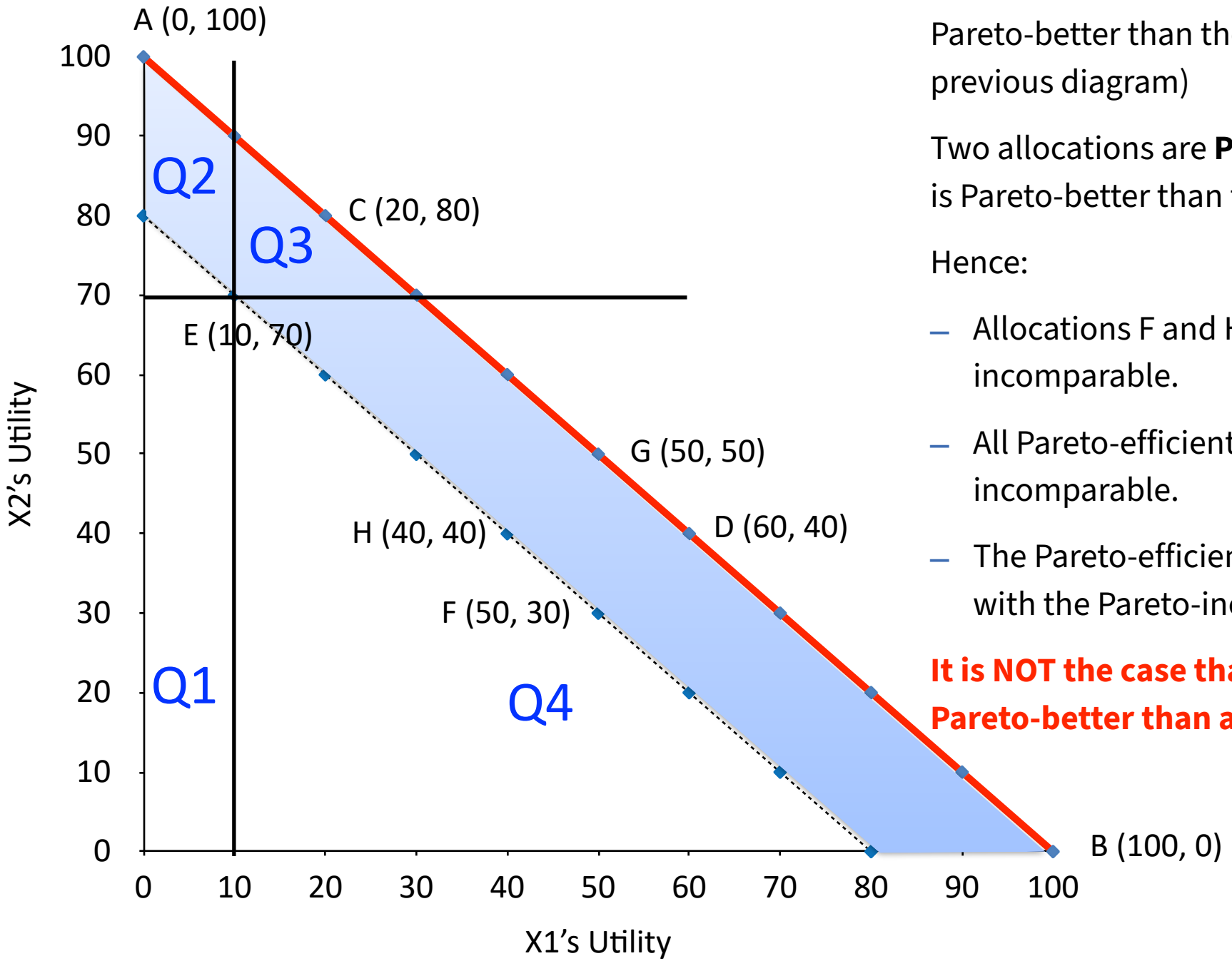
A policy is efficient if it is impossible to choose an alternative policy so as to make at least one person feel better off without making someone else feel worse off.



A distribution x is **Pareto-better** to a distribution y if:

1. each individual values x at least as much as y, and
2. at least one individual has a strict preference for x.

A distribution is **Pareto-efficient** if no Pareto-better distribution is available.



Two allocations are **Pareto-comparable** if one allocation is Pareto-better than the other (quadrants Q1 and Q3 in previous diagram)

Two allocations are **Pareto-incomparable** if no allocation is Pareto-better than the other (quadrants Q2 and Q4)

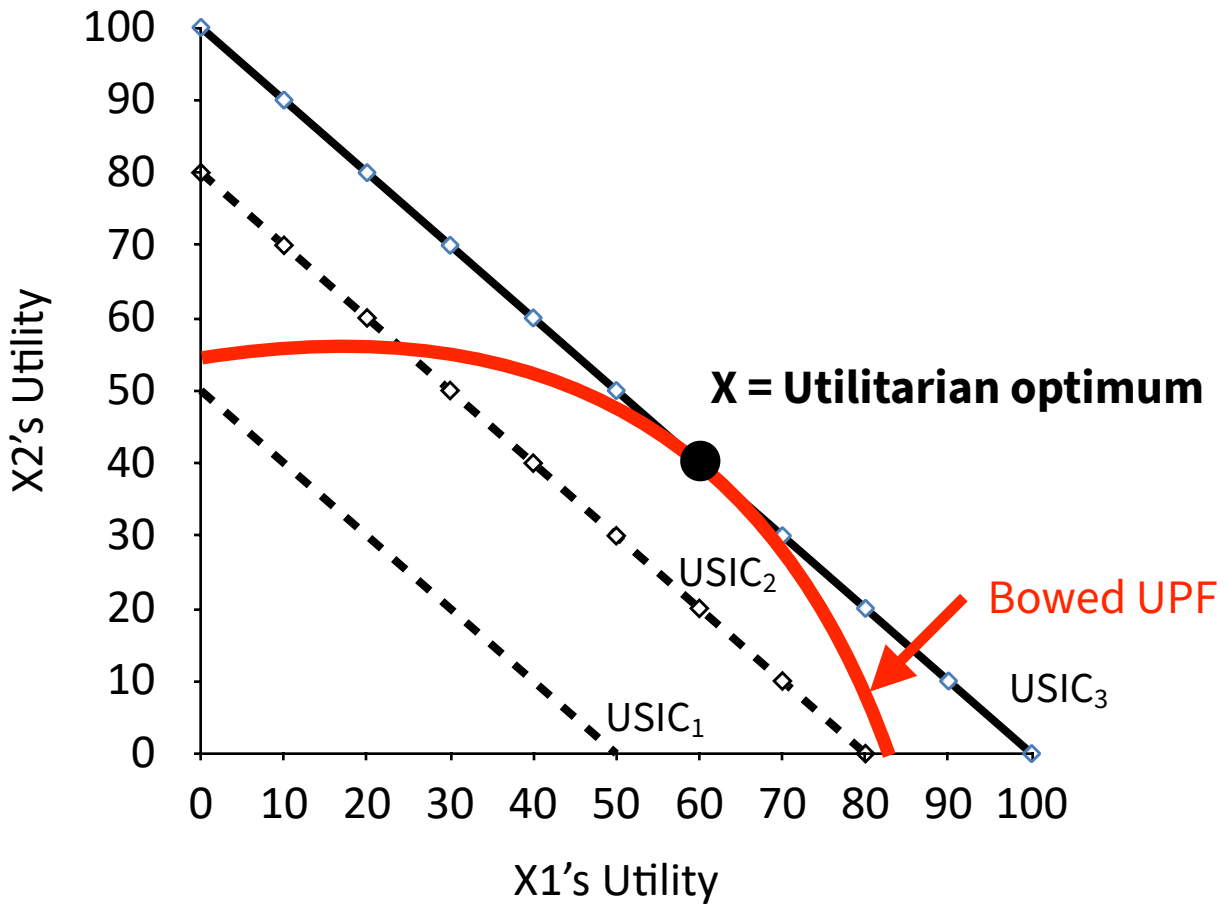
Hence:

- Allocations F and H, F and E, H and E are Pareto-incomparable.
- All Pareto-efficient allocations (e.g. D and G) are Pareto-incomparable.
- The Pareto-efficient allocation D is Pareto-incomparable with the Pareto-inefficient allocation E.

It is NOT the case that all Pareto-efficient allocations are Pareto-better than all Pareto-inefficient allocations!

The Relationship between Utilitarianism and Pareto

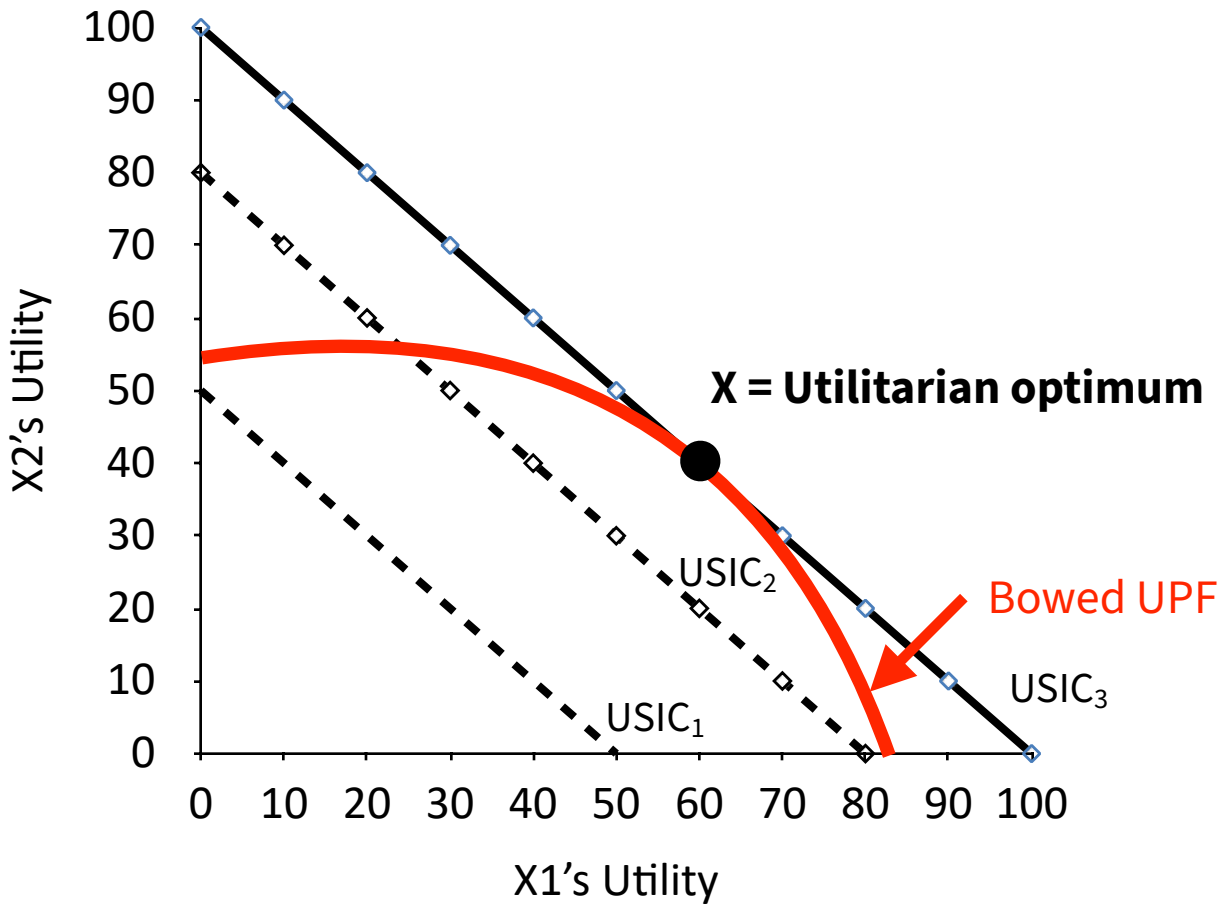
‘Strong’ vs. ‘weak’ welfarism



Utilitarianism respects Pareto: If someone gains and no one loses, total happiness rises. So every Pareto improvement is also a utilitarian improvement; and any utilitarian optimum is Pareto efficient!

The Relationship between Utilitarianism and Pareto

'Strong' vs. 'weak' welfarism



Utilitarianism respects Pareto: If someone gains and no one loses, total happiness rises. So every Pareto improvement is also a utilitarian improvement; and any utilitarian optimum is Pareto efficient!

But the converse fails!

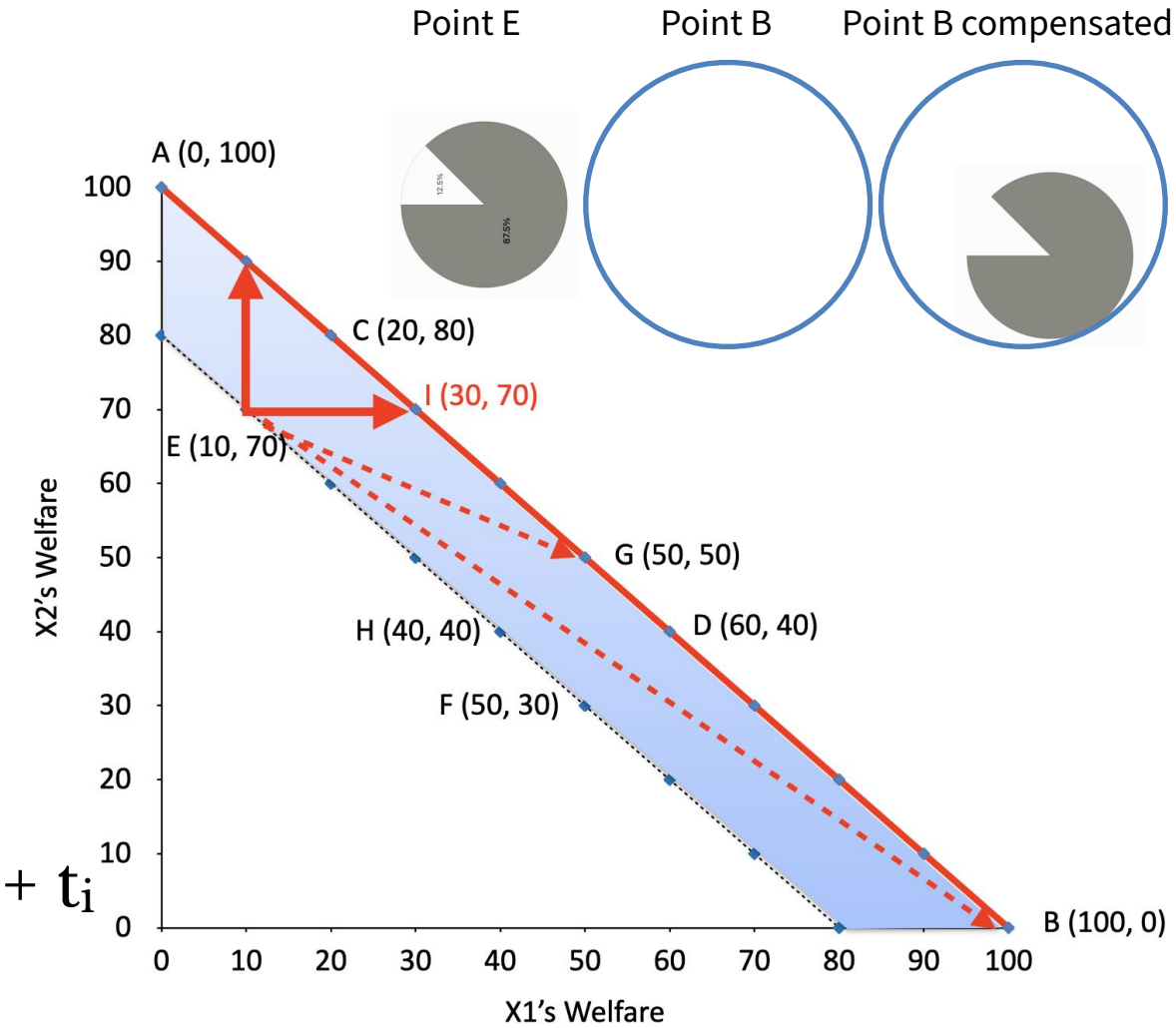
- The number of Pareto-efficient allocations is ∞. Pareto can't choose among them. Utilitarianism can.
- Utilitarianism endorses many changes that aren't Pareto-improvements. But almost every real-world policy has winners and losers. So the Pareto framework is almost always too demanding!

Pareto efficiency is a necessary condition for a utilitarian optimum, not a sufficient one!

Making Pareto Relevant for Public Policy

Kaldor-Hicks efficiency / potential Pareto improvements

- How should one evaluate a policy that brings society (say) from point E to point B?
- Suggestion by Nicholas Kaldor & Nicholas Hicks (1939): could the gainers compensate the losers in cash?
- If the "compensation test" is successful, we have achieved a potential Pareto improvement!
- Key premise: compensation is possible when a growing 'utility pie' is also a growing 'money pie'.
- This condition holds under the (heroic) assumptions of:
  - **Quasi-linear utility functions.** $U_i(a, t) = v_i(a) + t_i$
  - **Possibility of costless lump-sum transfers.**



Keeping Efficiency and Equity Separate

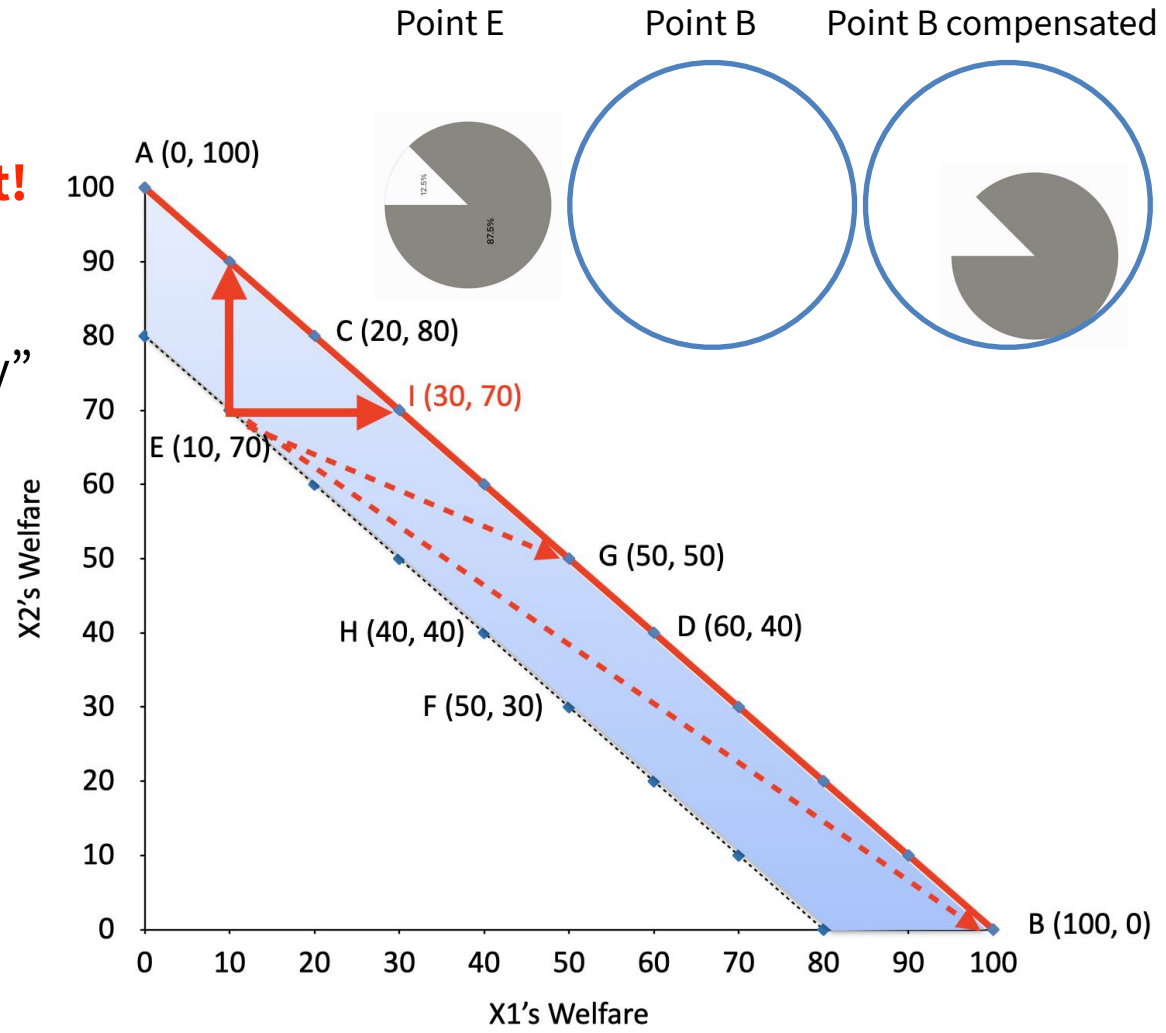
A lesson learnt from neo-classical economics

First Welfare Theorem

Under ideal assumptions, competitive markets will always lead to a Pareto-efficient outcome.

Efficiency first!

"Choose the large-pie policy"



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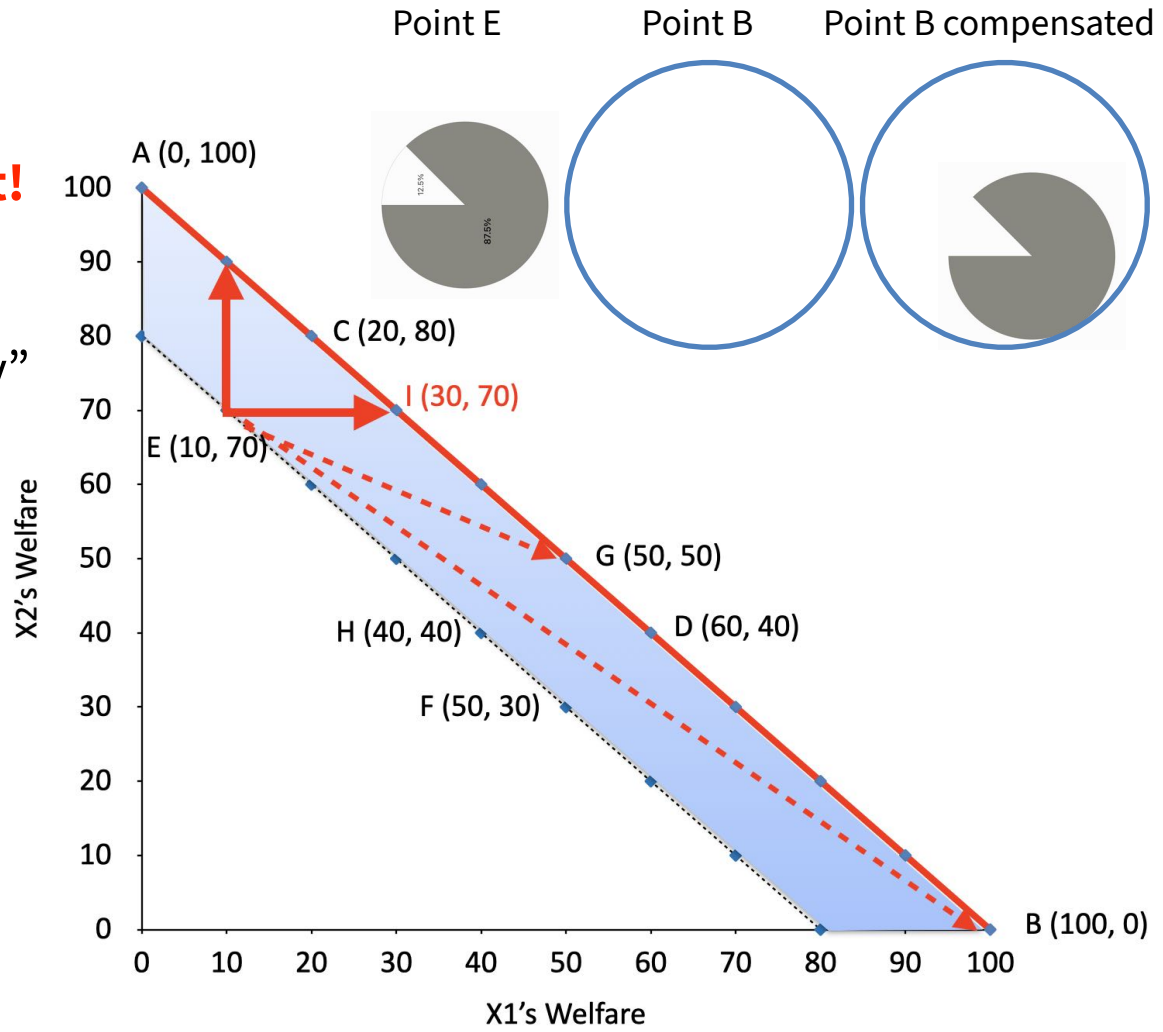
"Choose the large-pie policy"

Second Welfare Theorem

Society can attain any Pareto-efficient outcome by redistributing initial wealth among individuals and then allowing them to freely trade.

Redistribution second!

"Use transfers to reach the distribution you want"

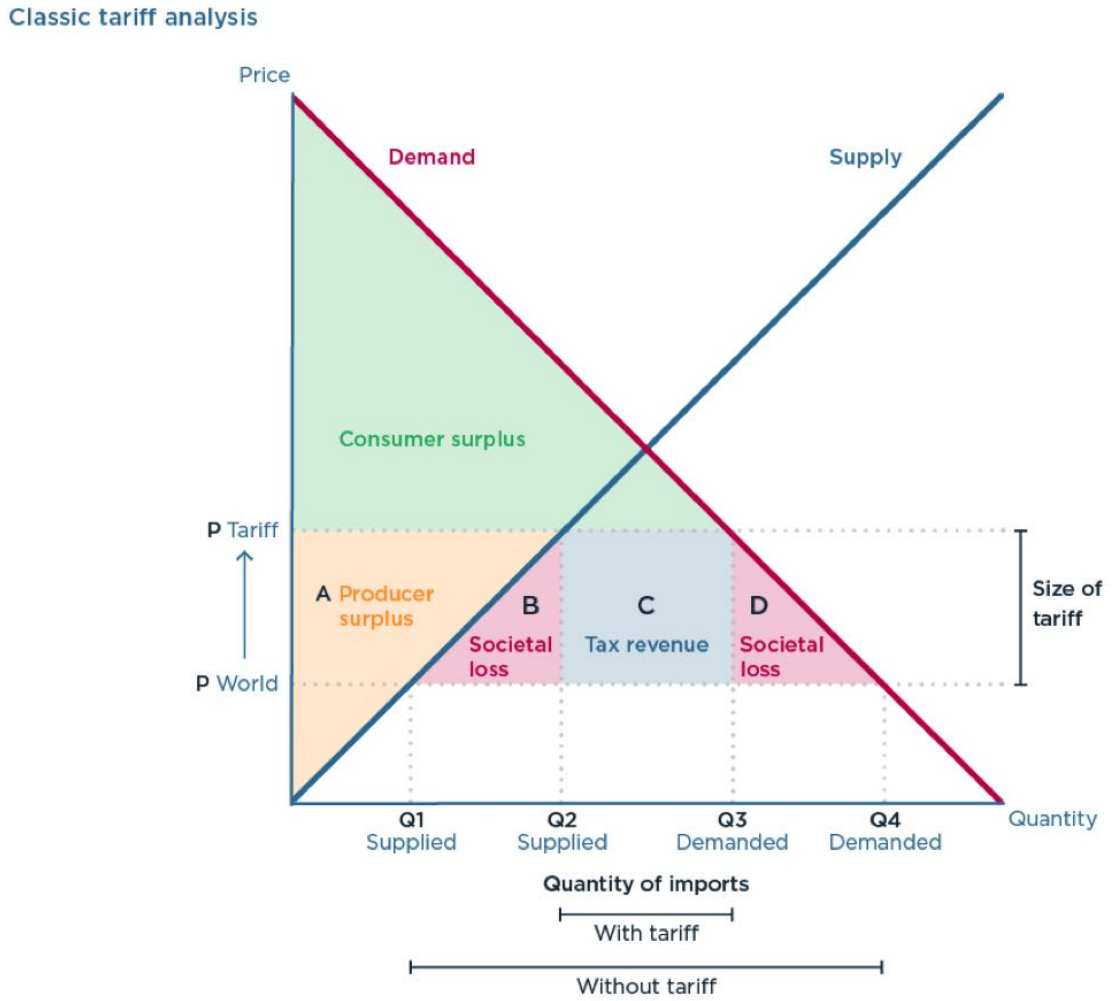


What If We Choose the “Small-Pie” Policy?



Suppose: $W = U_{Consumers} + U_{Producers}$

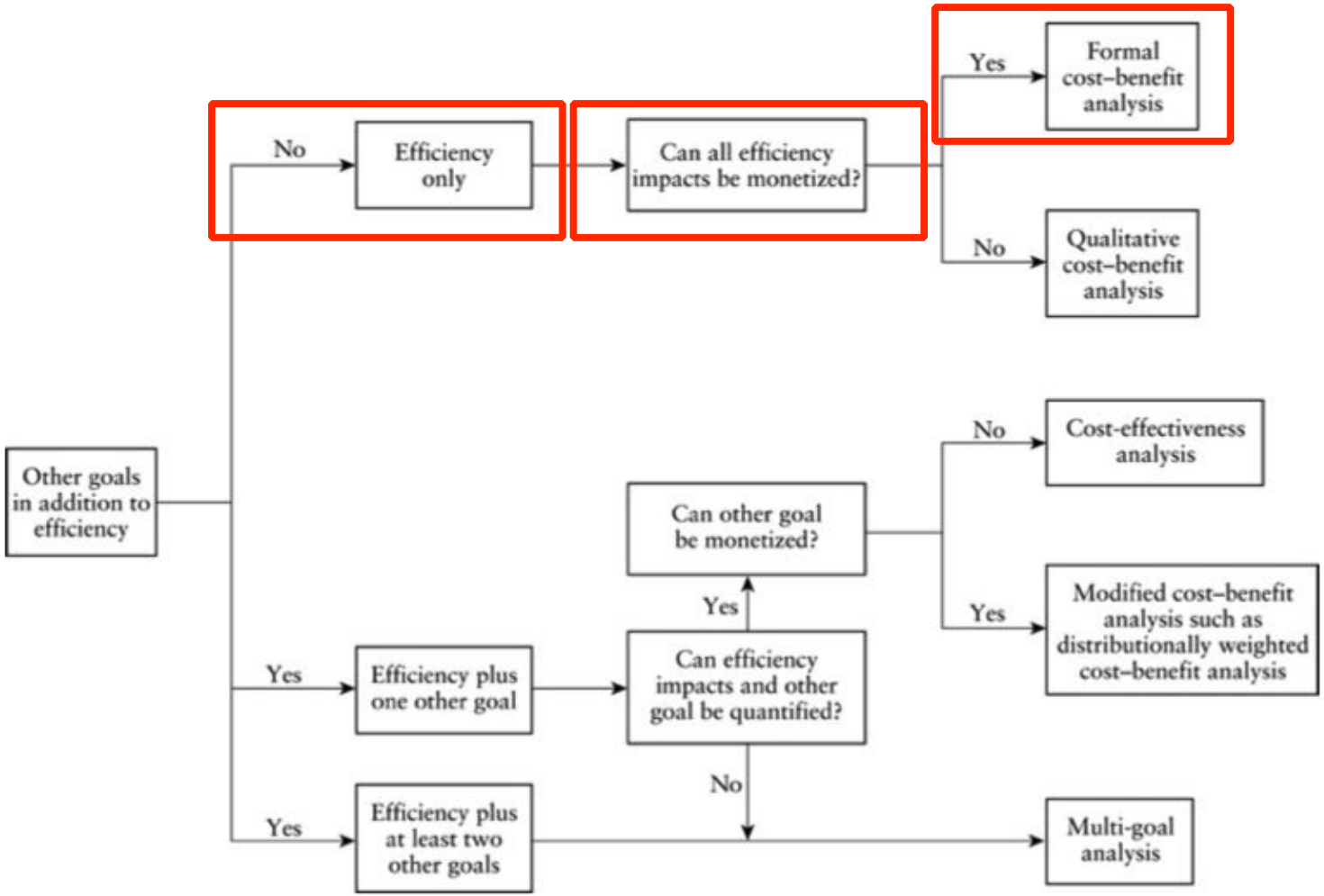
Analyse the welfare impact of tariffs in the U.S.!



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CBA: Identifying Potential Pareto Improvements in Practice

"Is policy project P efficient?"



Weimer & Vining 2017, Ch. 15

- What policymakers need: a response based on "actionable," $-quantifiable evidence.
- Cost-benefit analysis, generic form:

  Net Present Value (NPV) of P =

  present value of P's **Benefits**

  – present value of P's **Costs**.
- If the benefits flowing from a proposed project P exceed the costs occasioned by P, P should be undertaken!
- A positive NPV indicates a potential for "growing the utility pie" *and* a potential that losers be compensated.

Example: NPV of Non-Pharmaceutical Interventions (NPIs)

Value of a statistical life

Deaths without & with social distancing

Planning horizon

GDP forecast with & without SD

Social discount rate

$$\text{NPV} = \text{VSL} \times (D_1 - D_2) - \sum_{t=0}^{T} (Y_{1t} - Y_{2t})(1+r)^{-t}$$

Benefits (lives saved)

Costs (present value of GDP lost)

From Linda Thunström et al. 2020. "The Benefits and Costs of Using Social Distancing to Flatten the Curve for COVID-19." *Journal of Benefit-Cost Analysis* 11(2), 179-195

Example: NPV of Non-Pharmaceutical Interventions (NPIs)

NPV of a policy mix reducing the contact rate within the U.S. population in Spring 2020 by 38%.

Table 1 Benchmark outcomes for the uncontrolled scenario (without social distancing) and controlled scenario (with social distancing).

	Uncontrolled	Controlled
Infections (million)	287	188
Deaths (million)	2.18	0.941
Present value of GDP loss (trillion US$)	6.49	13.7
Value of lives lost (trillion US$)	21.8	9.41
Net benefits (trillion US$)		5.16

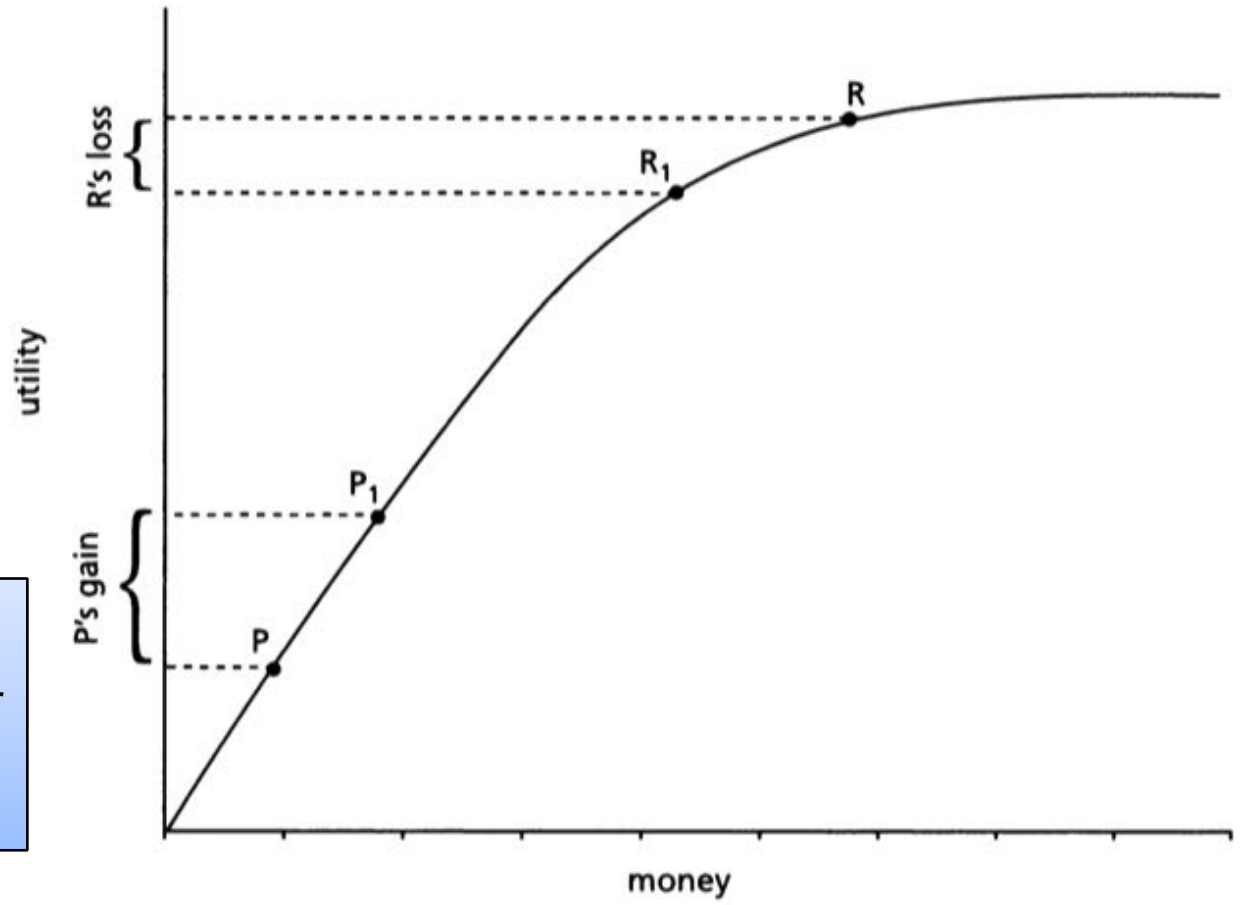
Benchmark parameter values: $R_0 = 2.4$, infectious period = 6.5 days, low mortality rate = 0.5 %, high mortality rate = 1.5 %, reduction in contact rate = 38 %, VSL = $10 million, uncontrolled initial GDP decline = 2.0 %, controlled initial GDP decline = 6.2 %, medical capacity threshold = 36 million infected, equal proportional rates of recovery in uncontrolled and controlled scenario, discount rate = 3.0 % $year^{-1}$, planning horizon = 30 years.

But The Separation Between Utility and Money May Fall Apart!

If a Euro is worth more in utility to the winners than to the losers of a policy, a policy change can raise total utility while the winners' willingness to pay (WTP) is smaller than the losers' willingness to accept (WTA).

Compensation is impossible!

Assuming diminishing marginal utility of income, wealthier people typically have higher WTP/WTA for a given utility change than poorer people do.



Thank You